Models Harmony cannot run

Equations of the PowerImpedance 0.4.0 test models that have no Harmony counterpart, and the examples and tests those equations block. Power is positive into the PowerImpedance component. Electrical states of converters and machines are per unit on the device bases. w_b is the base electrical frequency in rad/s.

Which cases these equations block

Harmony already matches the grid-following MMC (PLL, active power or DC voltage, reactive power, output-current control, circulating-current suppression, uncompensated modulation), Y-Y transformers, overhead lines, and cables. The rows below are the studies that stop on a missing device. The Frobenius numbers are PowerImpedance against its own PSCAD or Z-tool files, not against Harmony.

Case	Samples	Mean	What is missing
GFM MMC, total energy, uncompensated	58	5.7e-4	Virtual swing equation and virtual impedance
GFM MMC, leg energy, compensated	192	5.9e-3	Leg energy and compensated insertion
Two-level converter, Vdc and QV lag	28	1.5e-3	Two-level converter
Two-level converter, P and Q	36	1.6e-3	Two-level converter
Synchronous machine	240	3.4e-3	Field, dampers, AVR, governor
Induction machine	80	9.4e-4	Inverse-Gamma induction machine
Energy hub as a system	—	—	Offshore two-level converter
IEEE 39-bus STATCOM	—	—	Two-level converter
SMIB and converter-machine study	—	—	Synchronous machine
GFM weak grid	—	—	Swing equation, virtual impedance, two-level farms
Four-port bipolar MMC scan	192	—	No tool emits d, q, positive, negative

The synchronous-machine maximum relative error is 0.20 in one frequency bin. The package test still passes because it applies a relative tolerance of 9e-3 to the whole sweep at once.

1. Two-level converter

Reactor inductance L and resistance R, modulation m, DC voltage v_{dc}. Used by the energy-hub wind farm (700 MW, -150 Mvar), the three wind farms in the weak-grid example, the IEEE 39-bus STATCOM, and adm_TLC_test.

$$v_{Md} = v_{dc}\, m_d / 2, \quad v_{Mq} = v_{dc}\, m_q / 2$$

$$di_d/dt = (w_b / L)\,(v_{Md} - v_{Gd} - R\, i_d - L\, i_q)$$

$$di_q/dt = (w_b / L)\,(v_{Mq} - v_{Gq} - R\, i_q + L\, i_d)$$

$$i_{dc} = (v_{Md}\, i_d + v_{Mq}\, i_q) / v_{dc}$$

The validated controls are a PLL and an inner current PI, plus either DC-voltage control with a first-order QV lag, or active-power control with a first-order frequency lag. Harmony has no two-level bridge.

2. Grid-forming synchronization and virtual impedance

Replaces the PLL-as-frame used by the grid-following MMC. Harmony's grid-forming path is a filtered P and Q droop with an optional virtual R and L, and it cannot run together with the output-current loop. These two blocks are what GFM_weak_grid.jl and adm_MMC_test use.

Virtual swing equation. H is the virtual inertia in seconds. K_d damps against a PLL estimate of grid frequency. K_w is the frequency droop.

$$2\,H\, dw/dt = p_{ref} - p_f - K_d\,(w - w_{PLL}) - K_w\,(w - w_{ref})$$

$$d(theta)/dt = w_b\,(w - 1)$$

Current-controlled virtual impedance. A voltage error is turned into current references through $R_v + j\, w\, L_v$. The output-current loop then tracks those references.

$$i_d^{ref} = (R_v\, dv_d - w\, L_v\, dv_q) / (R_v^{\,2} + (w\, L_v)^2)$$

$$i_q^{ref} = (R_v\, dv_q + w\, L_v\, dv_d) / (R_v^2 + (w\, L_v)^2)$$

with $dv_d = V_d^{ref} + v_{outer} - v_d$ and $dv_q = V_q^{ref} - v_q$. The uncompensated total-energy loop that rides on this control is a single PI on the summed arm energy W, with power feedforward:

$$i_{sum,z}^{ref} = (K_p\, (W^{ref} - W) + xi + P_f) / (3\, v_{dc})$$

3. Leg energy and compensated modulation

Used by adm_MMC_compensated_test. Sum-energy and delta-energy each pass a notch at twice the fundamental and a rotating-frame PI, and they produce circulating-current references in d, q, and zero sequence. Harmony's energy loop is one total-sum channel and has no 2 w notch.

Compensated modulation does not apply the voltage command as an insertion index. It inverts the map from the seven insertion indices (sum d, q, z and delta d, q, z) to the arm voltages, using the instantaneous capacitor voltages. The command vector is [$v_{sum,d}$, $v_{sum,q}$, $v_{sum,z}$, $v_{delta,d}$, $v_{delta,q}$, 0, 0], and the 7 by 7 matrix is built from $v_{C,sum}$ and $v_{C,delta}$ in d, q, and zero sequence. Each path can carry its own Pade delay. Uncompensated modulation, which Harmony does have, skips that inversion and uses $m = 2\, v^{ref} / v_{dc}$.

4. Synchronous machine

Getting_started_SMIB.jl, Converter_machine_interaction.jl, and adm_SM_test. One field winding and one d-axis damper, two q-axis dampers, stator leakage L_l, and an external transformer L_t, R_t. Harmony's generator is a single per-phase field winding with R_f, T_f, and X_d.

$$L_d = L_{ad} + L_l + L_t, \quad L_q = L_{aq} + L_l + L_t$$

$$psi_d = -L_d\, i_d + L_{ad}\, (i_f + i_{d1}), \quad psi_q = -L_q\, i_q + L_{aq}\, (i_{q1} + i_{q2})$$

$$T_e = -(psi_d\, i_q - psi_q\, i_d)$$

Stator, field, and damper currents are the states. Their derivatives solve $L_D\, di_{d,group}/dt = w_b\, (v - R\, i - w\, psi)$ and the same on the q axis, with the speed-voltage sign flipped between the axes. The inductance matrices couple the stator, the field, and the dampers through L_{ad} and L_{aq}.

Rotor and steam turbine.

$$dw/dt = (F_{HP}\, T_{HP} + F_{IP}\, T_{IP} + F_{LP}\, T_{LP} - T_e) / (2\, H)$$

$$d(delta)/dt = w_b\, (w - 1)$$

Governor, with droop R and servo lags T_w, T_G.

$$d(dw_f)/dt = (1 - w - dw_f) / T_w$$

$$df/dt = (dw_f / R + T_m - f) / T_G$$

AVR. Terminal voltage is reconstructed behind the transformer impedance, filtered with T_R, passed through a lead-lag (T_C, T_B), and applied to the field with gain K_A and exciter lag T_A.

$$dv_f/dt = (sqrt(v_{td}^2 + v_{tq}^2) - v_f) / T_R$$

$$y = (T_C / T_B)\, (v^{ref} - v_f) + (1 - T_C / T_B)\, xi$$

$$dv_{df}/dt = (K_A\, y\, R_f / L_{ad} - v_{df}) / T_A$$

5. Induction machine

adm_IM_test. Inverse-Gamma model in a synchronous frame whose internal q axis leads. The external port uses a lagging q axis, and the interface changes the sign. Harmony's type-3 model is a DFIG with rotor and grid converters, not this machine.

$$L_M = L_m^2 / (L_m + L_{rl}), \quad L_{sigma} = L_m + L_{sl} + L_t - L_M$$

$$R_R = (L_m / (L_m + L_{rl}))^2\, R_r$$

$$di_d/dt = (w_b / L_{sigma})\, [v_d - (R_s + R_R)\, i_d + L_{sigma}\, i_q + w_r\, psi_q + (R_R / L_M)\, psi_d]$$

$$di_q/dt = (w_b / L_{sigma}) [v_q - (R_s + R_R) i_q - L_{sigma} i_d - w_r psi_d + (R_R / L_M) psi_q]$$

$$d psi_d/dt = w_b [R_R i_d - (R_R / L_M) psi_d + (1 - w_r) psi_q]$$

$$d psi_q/dt = w_b [R_R i_q - (R_R / L_M) psi_q - (1 - w_r) psi_d]$$

$$T_e = psi_d i_q - psi_q i_d$$

The mechanical load is $T(w) = T_0 (A w^m + B w + C)$, and the speed state is $dw_r/dt = (T_e - T(w_r)) / (2 H)$.

6. Independent-pole bipolar MMC

data_MMC_bipolar.txt is a 192-frequency scan with sixteen entries, ports d, q, positive, and negative. Neither PowerImpedance 0.4.0 nor Harmony produces it. Both MMC small-signal models have one DC channel. The energy-hub cables use two conductor positions and are then tied to that single DC port, so the hub study is a symmetric plus-minus link, not two independent poles.

Source of the equations: PowerImpedance.jl 0.4.0, ElectricalTLC, VSEWithDamping, CCVI, TotalEnergyControl, CompensatedModulation, ElectricalSM, MechanicalSM, GovernorSM, AVRSM, and ElectricalIM.